where the half-lives are given in years (y) and days (d)[2]. The three naturally occurring radioactive decay series, which are known as the *thorium series*, the *uranium series*, and the *actinium series*, are shown in Fig. 1.2. A fourth series, which originates in the synthetic element neptunium, is also shown. This series is not found naturally on earth since all of the radioactive species have decayed away long ago. Both the present symbolism of the isotope as well as the historical (i.e. "radioelement") symbolism are given in Fig. 1.2. Note that the *rule of Fajans and Soddy* is followed in each series so that α-decay causes a decrease in atomic number by two units and mass number by four, whereas β-decay produces no change in mass number but an increase in atomic number by one unit. Moreover, we see a pattern occurring frequently in these series where an α-decay step is followed by two β-decay steps. All known isotopes of elements $_{81}$Tl to $_{92}$U are given in Figure 13.4.

1.4. ATOMIC MODELS

Neither radioactive decay nor the discovery of isotopes provided information on the internal structure of atoms. Such information was obtained from scattering experiments in which a substance, such as a thin metal foil, was irradiated with a beam of ⟨-particles and the intensity (measured by counting scintillations from the scattered particles hitting a fluorescent screen) of the particles scattered at different angles measured (see Fig. 10.4). It was assumed that the deflection of the particles was caused by collisions with the atoms of the irradiated material. About one in 8000 of the á-particles was strongly deflected through angles greater than 90°. Consideration of these rare events led Rutherford in 1911 to the conclusion that the entire positive charge of an atom must be concentrated in a very small volume whose diameter is about 10^{-14} m. This small part of the atom he called the *nucleus*. The atomic electrons have much smaller mass and were assumed to surround the nucleus. The total atom with the external electrons had a radius of approximately 10^{-10} m in contrast to the much smaller radius calculated for the nucleus.

It was soon shown that the positive charge of the atomic nucleus was identical to the atomic number assigned to an element in the periodic system of Mendeleev. The conclusion, then, is that in a neutral atom the small, positively charged nucleus was surrounded by electrons whose number was equal to the total positive charge of the nucleus. In 1913 N. Bohr, using quantum mechanical concepts, proposed such a model of the atom which remains the basis of the modern atomic theory.

Our understanding of the nucleus has grown rapidly since Rutherford's scattering experiments. Some of the important steps in the history of nuclear science are listed in

[2] IUPAC recommends a for *annum*, instead of y, however y will be used throughout this text as it remains the commonly used term.

Table 1.1. Many of these discoveries and their practical consequences are discussed in the subsequent text.

1.5. NUCLEAR POWER

The peaceful use of the nuclear fission energy to produce electricity was first successfully employed in 1951 by the EBR-1 (Experimental Breeder Reactor) in the US. This was a fast breeder reactor using enriched uranium as fuel and cooled by a NaK eutectic melt. Originally it produced enough electricity to power four 200 W light bulbs and subsequently enough to power its own building. In 1955 the EBR-1 suffered a partial melt down and was finally deactivated in 1964. Admittedly the EBR-I produced the first electricity it was not really connected to the grid. This was achieved by the nearby plant, BORAX-III, which in 1955 made the nearby city of Arco to be the first city solely powered by nuclear energy.

Similar activities were performed in the Soviet Union where an existing graphite moderated channel-type plutonium production reactor was modified for heat and electricity generation by introducing a heat exchange system. It became the world's first thermal neutron nuclear powered electricity generator in 1954. The reactor was called AM-1 (Atom Mirny — peaceful atom) with a capacity of 30 MW_{th} or 5 MW_e. AM-1 produced electricity until 1959 and was used as a research facility and for production of radioisotopes until 2002. As seen above the general construction was close to the later infamous Chernobyl reactor RBMK (reaktor bolshoi moshchnosty kanalny — high power channel reactor).

The pressurised water reactor (PWR) history start with the use in naval propulsion. The Mark 1 prototype naval reactor became critical in 1953 which led to the first nuclear-powered submarine, *USS Nautilus*, was launched in 1954. This reactor led to the building in 1957 of the 60 MWe Shippingport demonstration PWR reactor in Pennsylvania. This reactor operated until 1982.

A boiling water reactor (BWR) prototype, Vallecitos, was operated from 1957 to 1963 but the first electricity producing one, Dresden-1 of 250 MWe was started in 1960 and was operating until 1978. There reactors also used enriched uranium fuel.

In Canada a different route was selected. Instead of going through the uranium enrichment process they created the CANDU reactor using natural uranium oxide fuel in a heavy water moderated and normal water cooled channel type reactor. The first one was started in 1962.

After these initial steps essentially four reactor systems are in general use: natural water moderated and cooled (PWR/VVER and BWR), heavy water moderated and natural water cooled (CANDU), and graphite moderated and water cooled (RBMK).